

ARUNESWAR S

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PROFESSIONAL SUMMARY

Performance-driven Electrical and Electronics Engineering student with hands-on experience in Industrial Automation, Embedded Systems, and Real-Time Hardware Development. Experienced in industrial electrical control panel assembly, embedded firmware development, electrical troubleshooting, and hardware validation through academic projects and industrial internships. National Winner of Smart India Hackathon 2024 for developing an AI-driven myoelectric prosthetic arm using ESP32 and EMG signal processing. Passionate about industrial automation, power systems, embedded electronics, and building reliable engineering solutions for real-world applications.

CORE COMPETENCIES

Industrial Automation & Electrical Systems: Industrial Control Panels | PLC Fundamentals | Servo Drive Integration | Electrical Panel Wiring | Electrical Schematics | Terminal Block Wiring | Ferrule Crimping | Cable Management | Testing & Commissioning | Electrical Troubleshooting | Industrial Safety

Embedded Systems & Firmware: ESP32 | STM32 | Embedded C/C++ | Real-Time Firmware | GPIO | ADC | PWM | UART | I²C | SPI

Electrical Engineering: Protective Relay Logic | Power Distribution | Control Systems | Fault Diagnosis | Hardware Validation | Signal Processing | Sensor Integration

Testing & Documentation: Digital Multimeter | Oscilloscope | Signal Tracing | Test Reports | Validation Testing | Root Cause Analysis

INDUSTRIAL EXPERIENCE

Electrical Panel Production Intern	June 2026 – July 2026
<i>Tech7 Automation System India Pvt Ltd, Coimbatore</i>	
- Assisted in the production, assembly, wiring, and testing of HT and LT electrical control panels for industrial automation applications. - Worked with industrial electrical equipment including Air Circuit Breakers (ACB), Vacuum Circuit Breakers (VCB), Current Transformers (CT), Potential Transformers (PT), MCCBs, contactors, relays, and terminal blocks. - Performed wire measurement, cutting, ferrule crimping, cable identification, terminal block wiring, and panel interconnections according to electrical schematics. - Supported panel production, cable routing, quality inspection, and pre-dispatch electrical testing while following industrial standards and safety procedures. - Gained practical exposure to industrial automation systems, PLC-based control panels, servo drive integration, HT/LT switchgear, and electrical testing procedures. - Assisted engineers and technicians in troubleshooting, inspection, and production activities to ensure reliable panel performance.	
Embedded Systems Engineering Intern	June 2025 – July 2025
<i>Dot World Technologies, Coimbatore</i>	
- Diagnosed and resolved hardware-software interface issues across embedded electronic systems using systematic signal tracing and schematic verification. - Conducted functional validation and testing under operational conditions to improve system reliability and performance. - Performed preventive inspection of sensors, relays, actuators, and embedded hardware components. - Prepared technical reports, validation documents, and fault analysis records to support debugging and maintenance activities. - Collaborated with engineers to troubleshoot hardware issues and improve overall system performance.	

PROJECT EXPERIENCE

AI-Based Myoelectric Prosthetic Arm	National Winner – Smart India Hackathon 2024
<i>Core Technologies: ESP32 EMG Sensors Servo Actuators Embedded C/C++ Python</i>	
- Developed an AI-assisted prosthetic arm capable of recognizing EMG muscle signals and controlling servo motors in real time. - Designed embedded firmware for EMG signal acquisition, filtering, and gesture classification using ESP32. - Optimized hardware and software to reduce electrical noise and improve gesture recognition accuracy. - Successfully demonstrated the working prototype during Smart India Hackathon 2024 and secured the National Winner title.	
Offline Smart Power Line Fault Detection System	
<i>Core Technologies: ESP32 Current & Voltage Sensors Relay Logic Embedded C OLED Display</i>	
- Designed an embedded system to monitor electrical distribution lines and detect abnormal operating conditions. - Implemented overcurrent, undervoltage, and short-circuit detection with automatic relay-based fault isolation. - Conducted multiple fault simulation tests to validate system performance and optimize protection thresholds. - Improved fault response through automated fault logging and relay control mechanisms.	

ACHIEVEMENTS

- National Winner – Smart India Hackathon 2024 (Government of India)
- First Runner-Up – Technocognition 2025, Dayananda Sagar University

EDUCATION

Bachelor of Engineering (B.E.) Electrical and Electronics Engineering	2023 – 2027 (Expected)
SNS College of Technology, Coimbatore	CGPA: 7.29 / 10